

Language Models and Author Identification

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<https://github.com/amberteetsel/nlp-author-identification>

ABSTRACT

This report covers the process of implementing a simple language model to predict authorship of a given text sample. Specifically, the classifier discussed is able to predict whether a passage was written by Sir Arthur Conan Doyle (*The Lost World*) or by J.R.R. Tolkien (*The Hobbit*). Components of the model include a Byte-Pair Encoding tokenizer, N-gram language models (bigrams and trigrams), and a perplexity measure. The classifier achieved 100% accuracy on unseen holdout data and correctly classified 10/11 out-of-domain passages from completely different novels. Classification errors are concentrated in dialogue-heavy passages.

Keywords: language models, N-gram, byte-pair encoding, tokenization, author identification

1 TOKENIZATION

Implemented a BPE-based tokenizer using Hugging Face's `tokenizers` library.

1.1 Design Choices

- **Pre-tokenizer strategy:** Initially used `Whitespace()` but switched to `ByteLevel()` to fix a functional bug. With `Whitespace()`, the decoder inserted a space between every token regardless of whether a token was a continuation of the same word. This caused words to split into multiple subword pieces. For example, rare words such as “gregarious” decoded to `gre gar ious`, “herd” decoded to `her d`. In contrast, `ByteLevel()` encodes word-boundary information directly into tokens via a leading-space marker, which enabled accurate decoding.

- **Byte coverage:** Seeded the trainer with:

```
initial_alphabet = pre_tokenizers.ByteLevel.alphabet()
```

to guarantee every byte had a valid vocabulary entry. This prevented rare or unseen characters from mapping to `<unk>`.

- **Shared vocabulary:** Trained a single BPE vocabulary on the combined training text of both novels to ensure token IDs were directly comparable across the author-specific language models.
- **Vocabulary size:** Tested vocabulary sizes `{100, 250, 500, 1000, 5000, 10000, 20000}` and measured bigram perplexity on holdout data at each size.

1.2 Key Observations

Vocab sizes of 100 and 250 produced identical results because both fell below the mandatory floor imposed by the byte-alphabet (256 tokens) plus special tokens (about 260 total tokens). Both configurations collapsed to essentially unmerged byte-level tokenization.

1.3 Methodological Findings

Perplexity is not directly comparable across vocabulary sizes because token granularity changes what “per-token” means. With a smaller vocab size, tokens are closer to individual characters which are easier to predict, and thus yield artificially low perplexity. Larger vocab sizes produce tokens that are closer to whole words, a harder prediction task that yields higher perplexity. Simply minimizing perplexity over vocab size would select byte-level tokenization, which is not a particularly meaningful outcome. Ultimately, chose vocabulary size of 5000 as a reasonable middle ground. To truly optimize vocabulary size, future analysis can evaluate downstream classification accuracy instead of raw perplexity.

2 N-GRAM LANGUAGE MODELING

Implemented a `NGramModel` class to store unigram, bigram, and trigram counts collected from tokenized training sentences. Each sentence was wrapped with `<s>/</s>` tokens to avoid context spilling over sentence boundaries.

2.1 Smoothing

Implemented add-k smoothing for both bigram and trigram probability estimation:

$$P(w_n | context) = \frac{\text{count}(\text{context}, w_n) + k}{\text{count}(\text{context}) + k \cdot |V|}$$

2.2 Inference

The model class includes a `neg_log_prob` method to return the total negative log probability of a token sequence under a given model and k value.

2.3 Limitations

Sentence boundary detection was performed with regular expressions, splitting on `. / ! ?` followed by a space and a capital letter. This occasionally misfired on abbreviations (e.g., “Mr. Smith” would be treated as two sentences with “Mr.” treated as the end of a sentence).

3 PERPLEXITY

Implemented perplexity as the exponentiated per-token average negative log probability, computed over a holdout set (approx. last 10% of each novel). The split was performed *after* cleaning and *before* tokenizer training to avoid contaminating the vocabulary with held-out content.

3.1 Bigram vs. Trigram (k = 1.0)

The trigram model performed substantially worse than the bigram at default smoothing (approx. 3-3.5x higher perplexity).

Book	Bigram Perplexity	Trigram Perplexity
The Hobbit	915.25	3244.32
The Lost World	117.04	3574.92

Table 1. Add-1 smoothing.

3.2 K-sweep

Calculated bigram and trigram perplexity for both books using k values from 0.001 to 1.0.

k	Hobbit Bigram	Hobbit Trigram	Lost World Bigram	Lost World Trigram
1.0	915.25	3244.32	1171.04	3574.92
0.5	684.38	2809.16	879.95	3168.45
0.1	387.96	1947.10	488.79	2306.41
0.01	273.99	1283.29	326.04	1582.17
0.001	346.57	1237.18	409.20	1518.34
0.0001	628.70	2010.28	779.24	2457.48

Table 2. K-sweep results. Lowest perplexity for each combination **in bold**.

Both bigram and trigram perplexity exhibit a U-shaped curve over k but with different optimal values. Bigram’s optimum $k \approx 0.01$ while trigram’s optimum is roughly 10x smaller at $k \approx 0.001$. This is consistent with trigram contexts

being sparser than bigram contexts because there is a much larger number of distinct trigram contexts so each is seen fewer times in training. This makes trigram probability estimates more sensitive to over-smoothing.

Regardless of k , the bigram model substantially outperforms the trigram model. This led to selecting a bigram model with $k = 0.01$ as the final configuration for the author-identification classifier.

4 AUTHOR IDENTIFICATION

4.1 Holdout Testing

Trained two independent bigram models, one on Tolkien (*The Hobbit*'s 90% training split) and the other on Conan Doyle (*The Lost World*'s 90% training split), using the shared vocabulary and bigram configuration with $k = 0.001$. To classify an unseen passage, the model computes the perplexity of the excerpt under both author models and predicts the author whose model yields the lowest perplexity.

Passage Length (sentences)	Accuracy	Counts
1	90.50%	638/705
2	98.02%	346/353
3	100.00%	235/235
4	100.00%	142/142
5	100.00%	71/71

Table 3. Validation results, by passage length in sentences, on held-out data.

Accuracy rises with passage length and reaches 100% at 3 or more sentences. This suggests the model needs only a small amount of context to reliably distinguish each author's writing style, though single-sentence passages remain harder to classify.

4.2 Error Analysis

At a passage length of 1 sentence, misclassified Tolkien sentences were predominantly dialogue while misclassified Conan Doyle sentences were a mix of dialogue and narrative prose. Investigated two hypotheses to explain this asymmetry:

1. **Dialogue proportion differences (rejected).** The initial check using a straight double-quote character (") suggested the *The Hobbit* was only 1.0% dialogue versus *The Lost World*'s 20.1%. However, source text for *The Hobbit* used curly quotes (" / ") which were not detected. After correcting the search to include curly quotes, *The Hobbit* actually comprised 21.7% dialogue so dialogue proportion was ruled out as the explanation.
2. **Point-of-view structure (supported).** Measured the rate of first-person pronoun usage (I, me, my, mine, myself) within non-dialogue narration sentences. *The Hobbit* used first-person pronouns in only 9.1% of sentences versus 35.6% for *The Lost World*. This reflects the structure of each book: *The Hobbit* is written in third-person while *The Lost World* is narrated in first person. Because Conan Doyle's narration and dialogue are both largely first-person, there is comparatively little distinction between the two stylistically. In contrast, Tolkien's third-person narration is distinct from dialogue which is necessarily first- or second-person. The Tolkien model's training material was roughly 90% third-person narration so it has a harder time classifying dialogue. This is a plausible explanation for why Tolkien's errors are concentrated almost exclusively in dialogue while Conan Doyle's are not.

5 CHALLENGES AND KEY OBSERVATIONS SUMMARY

- **Decode correctness depends on pre-tokenizer/decoder pairing.** `Whitespace()` alone produces incorrect decoded text for any multi-token word.
- **Training/inference pre-processing must match exactly.** A whitespace-normalization step applied during training-data preparation but not applied to test strings caused literal newline characters to map to `<unk>`, corrupting the decoded spacing. Fixed by applying the same normalization step inside `encode()`.

- **Gutenberg front matter required multi-stage, iteratively refined cleaning.** Simply stripping content outside of the header and footer was insufficient. Title pages, tables of contents, and chapter headings needed custom regular expression logic to avoid including non-prose structural text in the training data.
- **Perplexity not comparable across differing vocab sizes.** Any metric normalized per token is only comparable when the definition of a token is held constant.
- **Final classifier is highly accurate on passages of 3 or more sentences.** It achieved 100% accuracy on held-out data of sufficient length, and was reasonably accurate (90.5%) on single sentences.

6 OUT-OF-DOMAIN VALIDATION

To further test the classifier, it was evaluated on passages from **different novels** by the same two authors: *The Fellowship of the Ring* and *The Return of the King* (Tolkien), and *The Adventures of Sherlock Holmes* and *The Valley of Fear* (Conan Doyle). None of these texts were used in training. This is a more stringent test than the holdout evaluation because it assesses whether the classifier learned something in general about each author's writing style rather than simply recognizing patterns from *The Hobbit* and *The Lost World*.

6.1 Results: 10/11 passages correctly classified.

The single misclassification was an excerpt from *The Valley of Fear* consisting almost entirely of dialogue and written in a distinct dialect (e.g. "D'ye think I'm ashamed of my own name?"). The classifier confidently misidentified this as belonging to Tolkien: Tolkien perplexity was 509.66 versus Conan Doyle perplexity of 1097.65.

6.2 Interpretation

Verified this was not a result of a quirk with tokenization of data-encoding. Encoding and decoding the passage through the BPE tokenizer resulted in exactly the original text, confirming the misclassification reflects the bigram model's learned statistics instead of a corrupted input.

The result is consistent with earlier findings (Section 4.2) that dialogue is harder for the classifier to predict than regular narrative prose. In addition to being almost entirely dialogue, the misclassified passage is written in a different dialect and narrative structure (working-class American, third-person) than *The Lost World* (formal Victorian dialect, first-person) which the Conan Doyle bigram model was trained on. This is a plausible contributing factor for misclassification.

7 USE OF PRE-EXISTING CODE & ARTIFICIAL INTELLIGENCE

All AI-generated code was carefully reviewed and validated before implementation.

- **BPE Implementation:** Used Hugging Face's `tokenizers` library.
- **Test Set Loading:** AI-generated code to parse format of test passages and load them.
- **Final Evaluation Call:** AI-generated code for `main` in `evaluate_test_set.py`.
- **Everything Else:** Original code developed with AI assistance (Claude, Anthropic) for design discussion, debugging, and code review.